HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering

CSE221: Computer Algorithms

Batch: 23rd

Semester: Spring 2024

Date: 25/09/2024

Final Examination

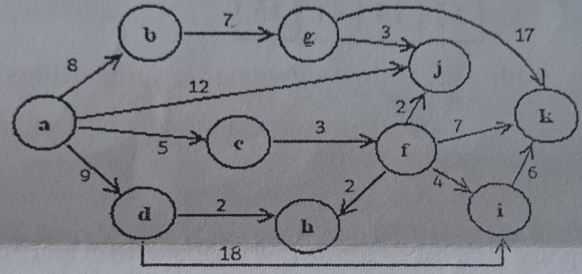
Time: 2 Hours

Total Marks: 40

Answer any **five** questions of the followings. $5 \times 8 = 40$

1. (a) State the approximation algorithm to store programs in disks. 3
   (b) Generate a heuristic algorithm to solve Knapsack Problem. 5

2. (a) You are given a directed weighted graph and the shortest path from a source vertex 's' to a destination vertex 'p'. If weight of every edge is increased by 5 units, does the shortest path remain same in the modified graph? Depend your opinion with an example. 3
   (b) Apply **Dijkstra's** single source shortest path algorithm to find the shortest paths with minimal cost from vertex '**a**' to other destinations of the following graph: 5



3. (a) Define Dynamic Programming. Write a pros and a cons of memorization in dynamic programming. 3
   (b) A system has three devices with the following costs and reliabilities: 5
   $\{c_1, c_2, c_3\} = \{30, 25, 15\}$ *and*
   $\{r_1, r_2, r_3\} = \{0.9, 0.75, 0.5\}$
   Use dynamic programming and design a three-stage system with a maximum cost of 115.

4. (a) Distinguish between Dynamic Programming and Recursion. 2
   (b) A city has four sell points P, Q, R and S with the following travel cost: 6

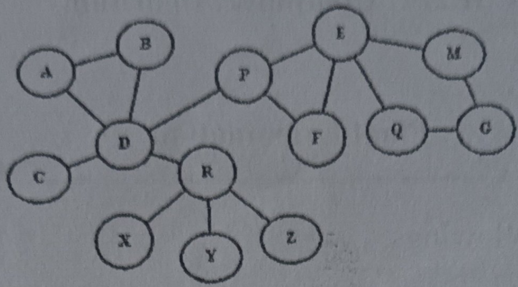
0	8	12	6
8	0	9	14
5	7	0	3
9	11	9	0

   Assume that, a salesperson wants to start the travel from P, cover all the sell points and return back to P. Find the optimal solution for him using dynamic programming.

5. (a) Compare between Algorithm and Decision Tree. 2
   (b) Write a short note on lower bound theory. 3
   (c) Prove that, any decision tree that can be sort n elements must have the height $\Omega(nlogn)$. 3

6. (a) Find two solutions for **n-queens** problem using backtracking when n = 4. 3

(b) Find the articulation points of the following graph with the help of **DFS** spanning tree. 5



7. (a) Apply branch-and-bound technique to solve 15-puzzle problem for the following organizations: 4

1	2	3	4
5	6	7	
9	10	12	8
13	14	11	15

(b) Let you have four jobs with the following penalties, deadlines and required times respectively: 4

$p_i = \{10, 8, 6, 5, 4\}$,

$d_i = \{2, 1, 3, 4, 2\}$ *and*

$\{t_i = \{1, 2, 1, 1, 2\}$

Now, generate the state space tree to sequence to jobs with deadlines using branch and bound approach.

HAMDARD UNIVERSITY BANGLADESH

Tamrin Hasan

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 223: Object Oriented Programming in Java

Batch: 23rd

Semester: Spring 2024

Date: 19/07/2024

Final Examination

Time: 2 hours

Total Marks: 40

Answer any **five** questions from the following: 5x8= 40

1 (a) How does inheritance promote code reuse? Provide a detailed explanation. 3

(b) Create classes Flyable and Swimmable with methods fly and swim respectively. Then, create a class Duck that inherits from both Flyable and Swimmable and implements both methods. 5

2 (a) Explain the concept of late binding in the context of run-time polymorphism. 3

(b) Create a class Complex for complex numbers. Overload the + operator to add two complex numbers and demonstrate compile-time polymorphism. 5

3 (a) Describe a scenario where you would use the private access modifier over the public access modifier. 4

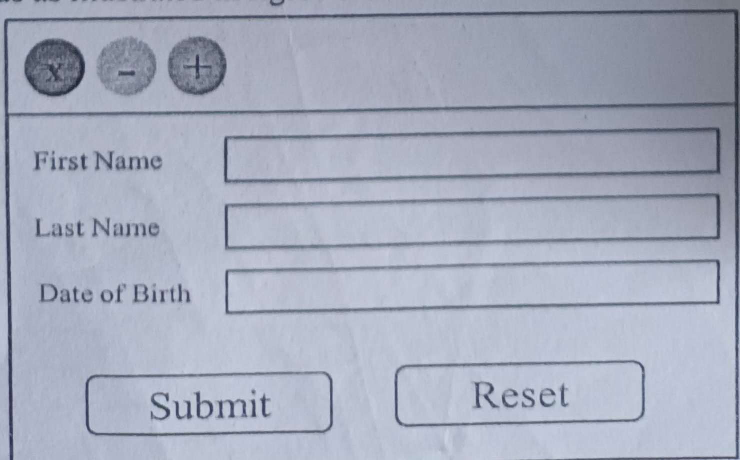
(b) Create a class Person with private, protected, and public fields and methods. Show how these fields and methods can be accessed from within the class, from a subclass, and from a different class in the same package. 4

4 (a) Differentiate between encapsulation and abstraction. 4

(b) Create a class Employee with private fields for name, age, and salary. Provide public getter and setter methods for these fields to implement encapsulation. 4

5 (a) Write the steps to perform event handling in AWT. 3

(b) Write a Java program using AWT to create a window with buttons and text fields as illustrated in figure below: 5



Tanvir Hasan

6 (a) How do interfaces support abstraction? 3

(c) Create an abstract class Vehicle with an abstract method startEngine. Then, create concrete classes Car and Bike that extend Vehicle and implement the startEngine method. 5

7 (a) Explain the advantages and disadvantages of native API driver. 4

(b) Find the output of the following Java program: 4

```
public class JavaMathExample
{
  public static void main(String[] args)
  {
    double a = 30;
    double b = Math.toRadians(a);
    System.out.println("Sine value of a is: " +Math.sin(a));
    System.out.println("Cosine value of a is: " +Math.acos(a));
    System.out.println("Tangent value of a is: " +Math.tanh(a));
    double x = 28;
    double y = 4;
    System.out.println("Square root of y is: " + Math.sqrt(y));
    System.out.println("Logarithm of y is: " + Math.log(y));
    System.out.println("log1p of x is: " +Math.log1p(x));
    System.out.println("log10 of x is: " + Math.log10(x));
    System.out.println("exp of a is: " +Math.exp(x));
  }
}
```

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

CSE 225: Digital Logic Design

Batch: 23rd

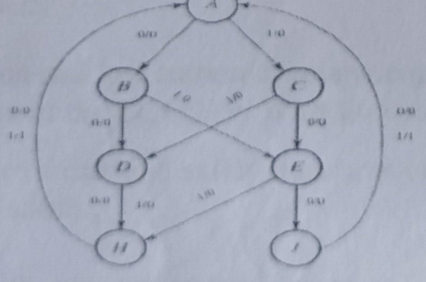
Semester: Spring 2024

Date: 05/10/2024 | **Final Examination** | Time: 2 Hours

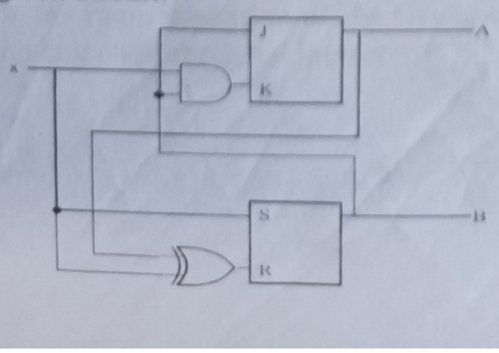
Total Marks: 40

Answer any **five** questions of the following. $5 \times 8 = 40$

1.	(a)	Draw the circuit diagram of an SR flip-flop and explain its operation with the help of a truth table.	CLO4	2
	(b)	What are the differences between racing and toggling in flip-flops? Provide an example to illustrate your answer.	CLO4	3
	(c)	Describe the operation of a T flip-flop and explain how it can be derived from a JK flip-flop.	CLO4 & CLO5	3
2.	(a)	You are given a JK flip-flop and need to convert it into a D flip-flop. Draw the circuit diagram.	CLO5	5
	(b)	Briefly explain why JK flip-flops are considered universal. Give one example of converting a JK flip-flop to another type.	CLO3	3
3.	(a)	Design a 3-bit synchronous up counter using JK flip-flops.	CLO5	5
	(b)	Explain the difference between a Moore machine and a Mealy machine.	CLO4	3
4.	(a)	Reduce the states from the following state diagram. Then draw the reduced state diagram.	CLO4	3



	(b)	Analyze the following circuit and draw the state diagram of the given circuit:	CLO4	5



5.	(a)	What is counter? Briefly discuss about the type of counter.	CLO4	2
	(b)	Make a comparative analysis between different type of counter.	CLO4	2
	(c)	Design a 4 bit UP-DOWN ripple counter, in the same circuit.	CLO5	4
6.	(a)	Design a counter to count the following arbitrary sequence using flipflops and basic gates. Sequence: $1 \rightarrow 0 \rightarrow 3 \rightarrow 6 \rightarrow 2 \rightarrow 5 \rightarrow 7 \rightarrow 1$. Here, the next sequence of the missing number 4 is 6. I. Draw the state Diagram. II. Find the state table with JK flip flop inputs. III. Minimize the functions of flip flop inputs. IV. Draw the circuit diagram using block Diagram of flip flops and basic gates.	CLO5	6
	(b)	Create a state diagram for a sequence detector that outputs a 1 when it detects the final bit in the serial data stream 1101.	CLO5	2
7.	(a)	Briefly explain the following terms. i) PLA, ii) PAL, iii) ROM.	CLO4	2
	(b)	Design a PAL for implementing the following equations. $A = \sum m(8, 9, 12, 13)$ $B = \sum m(4, 5, 12, 13)$ $C = \sum m(5, 7, 13, 15)$	CLO2	3
	(c)	Design a 64x4 ROM and explain it's working principle with an example.	CLO5	3

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology

Department of Computer Science and Engineering

ENS221: Environmental Science

Date: 28/09/24

Final Examination

Semester: Spring 2024
Time: 2 Hours
Total Marks: 40

Answer any **four (04)** of the following questions: $4\times10=40$

			CLO	
1.	a)	Define 'Environment Science'? What are the spheres of environment? Explain.	CLO1	5
	b)	What is the main idea of drought? Explain the features of it in the context of Bangladesh.	CLO2	5
2.	a)	Explain the process of arsenic contamination in Bangladesh's groundwater, and what measures can be taken to mitigate its impact on public health?	CLO2	4
	b)	What do you think regarding the major causes of flood in Bangladesh? Explain.	CLO2	6
3.	a)	Explain comprehensive disaster management.	CLO3	3
	b)	Write Six Pillars of Bangladesh Climate Change Strategy and Action Plan (BCCSAP) and explain the capacity building and institutional strengthening.	CLO3	7
4.	a)	Define Green Tourism. Identify the primary indications of it and elucidate.	CLO3	6
	b)	Give some suggestions for preventing the damage of flash flood damage in Bangladesh	CLO2	4
5.	a)	Mitigation and low carbon development is one of the most important pillar among six pillars of the BCCSAP. What do you think? Justify it.	CLO3	6
	b)	Write the threats and safety measures of lightning and thunderstorm in the context of Bangladesh.	CLO3	4
6.	a)	What is do you understand by waste? What are the key steps in effective waste management.	CLO3	5
	b)	How can the implementation of the 5Rs in waste management systems contribute to minimizing environmental impact?	CLO3	5

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 227: Data Communication

Batch: 23^{rd}
Date: 30/09/2024

Final Examination

Semester: Spring 2024
Time: 2 Hours
Total Marks : 40

Answer any **five** questions: $5 \times 8 = 40$

1. (a) Draw and describe amplitude modulation in analog-to-analog conversion. 4
   (b) Explain parallel and serial data transmission in your own words. 4

2. (a) List all categories of multiplexing. Define FDM. 3
   (b) Illustrate three strategies of time division multiplexing. 5

3. (a) Describe spared spectrum and its goal. 4
   (b) Draw and explain digital signal service (Digital hierarchy). 4

4. (a) How does sky propagation differ from line-of-sight propagation? Explain. 4
   (b) Write the differences between **omnidirectional** waves and **unidirectional** waves. 4

5. (a) Make a table for the distinct forms of UTP cables. 4
   (b) Write the advantages and disadvantages of fiber optic cable. 4

6. (a) Explain packet switching network. 4
   (b) Describe three phases of circuit-switched network. 4

7. (a) Distinguish between a point-to-point link and a broadcast link. 4
   (b) Illustrate Address Resolution Protocol (ARP) packet. 4

HAMDARD UNIVERSITY BANGLADESH

Faculty of Science, Engineering and Technology
Department of Computer Science and Engineering
CSE 226: Digital Logic Design Lab

Batch: 23rd

Semester: Fall 2024

Date: [redacted]

Final Examination

Time: 40 min

Total Marks: 40

Answer **All** of the questions: 40

1. What is a flip-flop? Explain the difference between SR and D flip-flops in terms of their inputs and behavior. 5

2. How does a synchronous counter differ from an asynchronous counter? Provide an example application for each type. 5

3. Design a 3-bit synchronous up-counter using T flip-flops. Draw the circuit diagram and explain its operation. 5

4. Create a 3-bit synchronous down-counter with an additional input 'x'. When x=0, the counter should count down normally, but when x=1, it should hold its current value. Use X flip-flops for your design and provide the circuit diagram. 25

Lab Test

Problem 1: Implement a 3-input XOR gate using only NAND gates.

Problem 2: Build a full adder circuit using Only Universal Gates.

Problem 3: Implement an SR flip-flop using NOR gates.

Problem 4: Design a circuit to display the digit "3" and "1" on a 7-segment display.

Problem 5: Convert a JK flip-flop to a T (Toggle) flip-flop.

Problem 6: Convert an SR flip-flop to a JK flip-flop.